

RiboFUSION: Supplementary Material

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This supplementary material accompanies the main text. It provides the complete per-method data, the component-level record, and the provenance note, so that every headline claim in the main text can be re-read against one frozen source.

S1. Frozen per-method metrics. Tables S1, S2, and S3 list every per-method metric for the three locked cohorts, read from a single frozen results table; no value is recomputed after locking. The RiboFUSION aggregate and its USDF-5 aggregate coincide on the internal cohort by construction of the locked value.

Table S1: Frozen per-method metrics, internal cohort (bold = RiboFUSION aggregate).

Method	AUC	AP	Brier	LogLoss	P@1	R@20	ECE
RiboFUSION	0.8663	0.7261	0.249	1.1287	0.9045	0.5485	0.259
USDF5	0.8663	0.7261	0.249	1.1287	0.9045	0.5485	0.259
USDF4	0.8722	0.7372	0.2262	1.8967	0.9361	0.5534	0.2254
RiboEncoder	0.8707	0.7263	0.147	0.4591	0.9182	0.5457	0.1179
MHCFlurry	0.8383	0.6734	0.1994	0.5941	0.8864	0.5152	0.2281
BigMHC	0.8379	0.6884	0.1416	0.5006	0.9045	0.53	0.0831
TransPHLA	0.8253	0.6283	0.2939	0.9621	0.7682	0.4958	0.3278
PLM-LR	0.7917	0.5724	0.1579	0.4866	0.8	0.4476	0.0395

Table S2: Frozen per-method metrics, Ouspenskaia cohort (bold = RiboFUSION aggregate). A dash marks a metric not computed for that method.

Method	AUC	AP	Brier	LogLoss	P@1	R@20	ECE
RiboFUSION	0.9207	0.9057	0.1529	1.7702	1.0	0.3794	0.1497
USDF5	0.914	0.9001	0.1629	0.4988	1.0	0.3776	0.1544
USDF4	0.9095	0.9006	0.1604	0.4948	1.0	0.378	0.1473
RiboEncoder	0.8451	0.8428	0.1608	0.4878	0.9935	0.3632	0.0175
MHCFlurry	0.8837	–	0.216	0.6337	0.987	–	–
BigMHC	0.9265	–	0.2082	–	0.9805	–	–
TransPHLA	0.8991	0.8648	0.1937	0.6565	0.9091	0.3564	0.2136
PLM-LR	0.8077	0.7781	0.1795	0.5433	0.8961	0.3337	0.0343

S2. Complete per-metric record. The per-metric values behind every figure and statement in the main text are collected in Table S4, so that the full record can be read together.

S3. Provenance and code. Every number in the paper is read from one frozen results table; the three provenance values in Fig. 4N are anchored against a 20k recompute, and the USDF-4 Ousp value is the live-recomputed 0.9095 (a stale 0.9096 present in an earlier official file is not

Table S3: Frozen per-method metrics, Chong cohort (bold = RiboFUSION aggregate).

Method	AUC	AP	Brier	LogLoss	P@1	R@20	ECE
RiboFUSION	0.9481	0.918	0.1131	1.6165	1.0	0.3589	0.087
USDF5	0.9438	0.9359	0.14	0.4376	0.875	0.3883	0.1714
USDF4	0.9422	0.9319	0.1339	0.4239	1.0	0.3883	0.1541
RiboEncoder	0.9043	0.8916	0.1312	0.4237	0.875	0.3792	0.082
MHCFlurry	0.9199	0.9065	0.1234	0.3983	0.875	0.3815	0.0963
BigMHC	0.9336	0.9304	0.1203	0.3893	1.0	0.3928	0.0899
TransPHLA	0.9238	0.899	0.1894	0.6324	1.0	0.3657	0.22
PLM-LR	0.8481	0.8178	0.1593	0.4891	0.875	0.3431	0.0498

Table S4: The complete per-metric record, with the section of the main text in which each value is used.

Metric	Value	Section
Component ladder AUROC (D1/N1/N2/N3/composite)	0.9000/0.9034/0.9052/0.9012/0.9058	Sec. 4.4
Per-allele AUROC range (rare bucket)	0.55–0.93	Sec. 4.4
Single-reader steps (N3 groove-transfer)	−0.0040	Sec. 4.6
Internal reader ladder (RiboEncoder vs composite)	0.8707 vs 0.8663	Sec. 4.7
Equal-weight fusion, internal AUROC	0.8575	Sec. 4.7
U4-to-U5 delta interval, Chong	crosses 0	Sec. 4.7
P@1 lower group (Chong)	0.875	Sec. 4.3
First-cohort AUC ordering (BigMHC vs RiboFUSION)	0.9265 vs 0.9207	Sec. 4.2
First-cohort per-sample AP (BigMHC vs RiboFUSION)	0.916 vs 0.9057	Sec. 4.2
Probability-calibration columns (Brier/LogLoss/ECE)	Table S1	Sec. 4.1
N2 gain by stratum (general / high-rare)	+0.0053 / +0.0025	Sec. 4.6
ncORF feature knockout (length removed)	−0.0047	Sec. 4.6
Entropy-confidence probe (partial corr., p)	0.005, $p = 0.49$	Sec. 4.8
IG-to-edit correlation	0.229	Sec. 4.8
Groove-transfer sweep (peak vs learned λ^*)	0.30 vs 0.62	Sec. 5.4

cited). The figure-generation scripts, the protocol lock, and the per-figure data files are released with the repository named in the data-availability statement below.